

DS5220 Homework 0

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Instructions

R has excellent visualization capabilities, especially with the `ggplot2` package. Please read Chapter 3 of *R for Data Science [GW]*, Garrett Grolmund, Hadley Wickham, and complete the exercises below after you finish each section. Edit the markdown file which came with this html directly. Make sure to enter your R code in the chunks following each question to demonstrate your answers. Follow each code block with a text description of your solution. Answers without demonstration will be given little credit. Code with no description (if requested) will be given little credit.

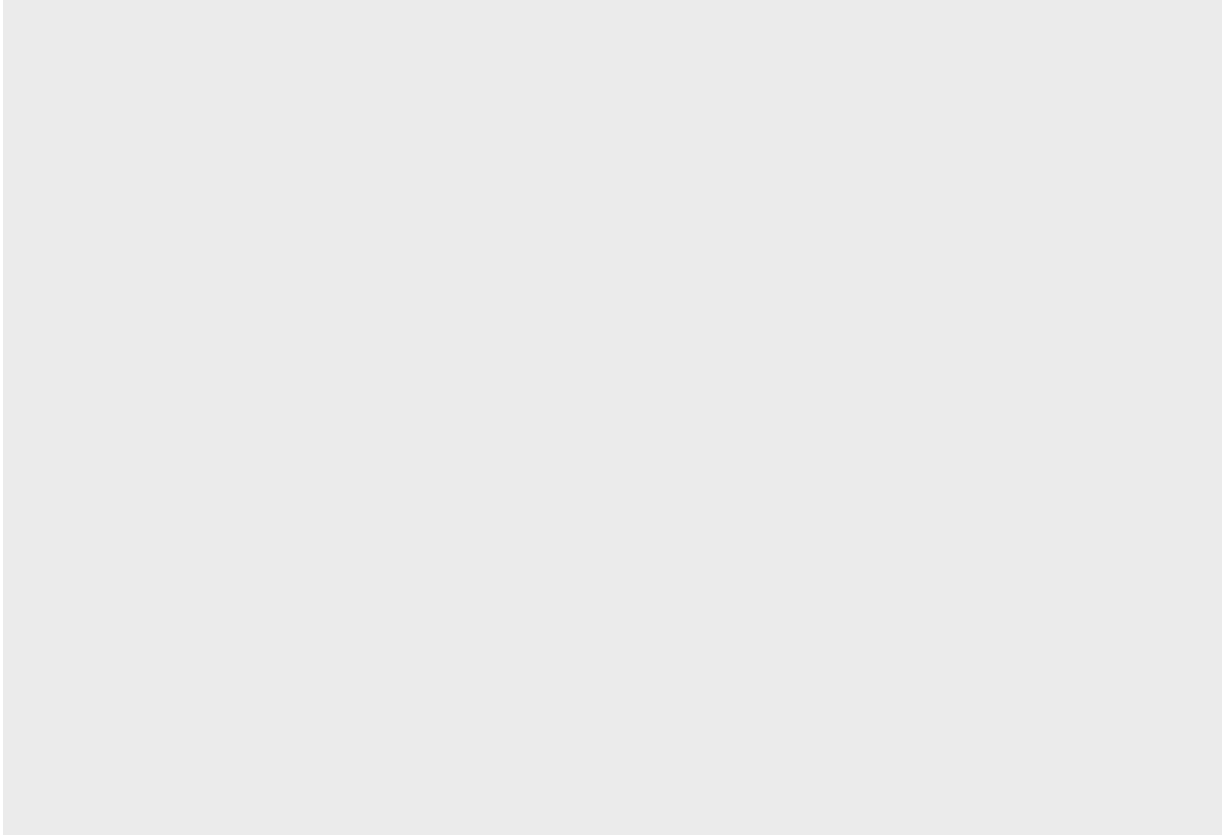
3.2.4 Exercises

1. Run `ggplot(data = mpg)`. What do you see?

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.3      v readr      2.1.4
## v forcats    1.0.0      v stringr   1.5.0
## v ggplot2    3.4.3      v tibble    3.2.1
## v lubridate  1.9.2      v tidyr     1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
ggplot(data = mpg)
```



When we run `ggplot(data = mpg)`, it initializes ggplot2 plotting system/package, but no specific plot is created yet because we haven't specified any mappings. This is why we see an empty plot which indicates that a plot is prepared but no data is given therefore no graph is shown.

2. How many rows are in `mpg`? How many columns? Demonstrate how you obtained your answers using R.

```
dim(mpg)
```

```
## [1] 234  11
```

```
#
```

I used the `dim()` function to check the dimensions of the dataset (I could have also used functions `nrow(mpg)` and `ncol(mpg)`). This will return the number of rows (234 in our case) and columns (11) in the dataset.

3. What does the `drv` variable describe? Read the Help Panel in RStudio by typing `?mpg` in the Console Panel to find out. (You will see no output from RMarkdown here.) Produce a description of `drv` by typing `mpg` below.

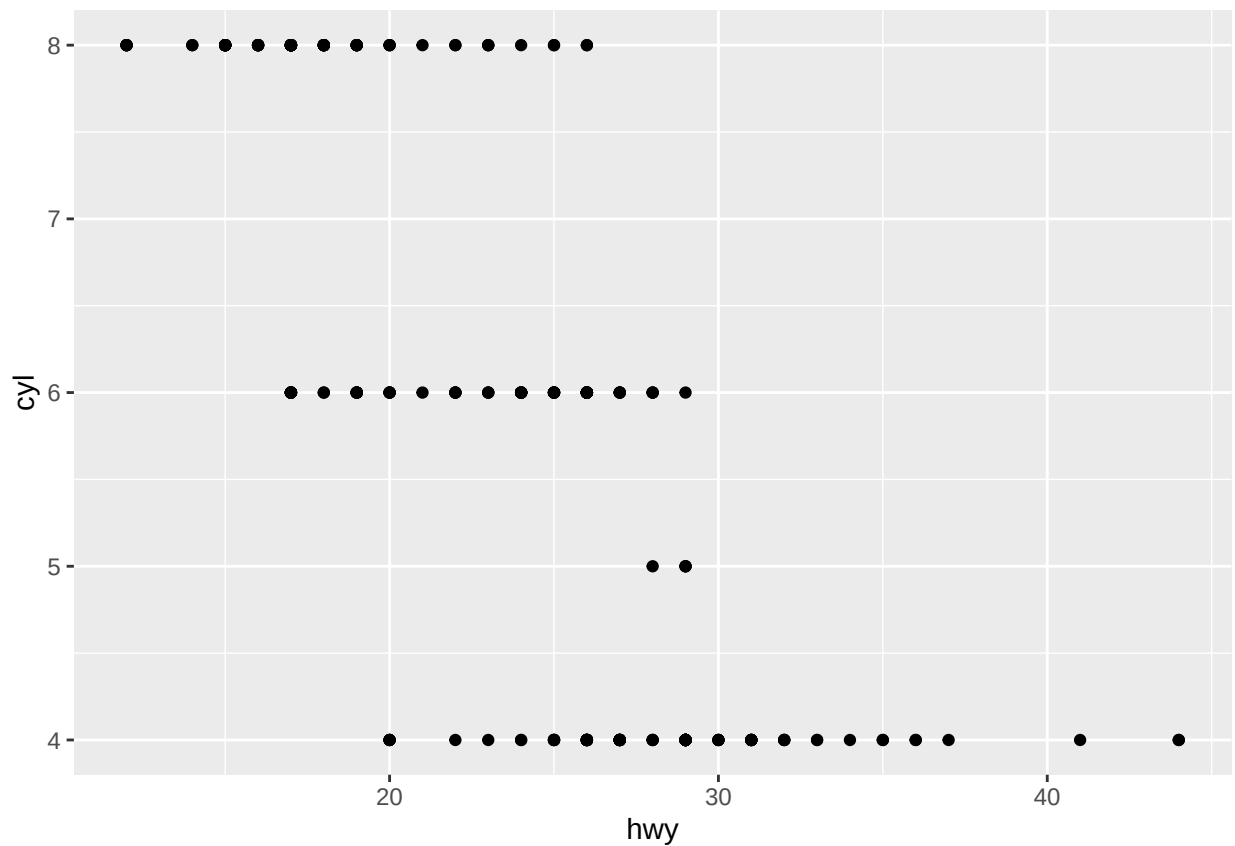
```
?mpg
```

```
## starting httpd help server ... done
```

The `drv` variable in the `mpg` dataset indicates the type of drive train in the vehicle. It classifies cars into front-wheel drive (f), rear wheel drive (r), and four-wheel drive (4).

4. Make a scatterplot of `hwy` vs `cyl` using `geom_point`.

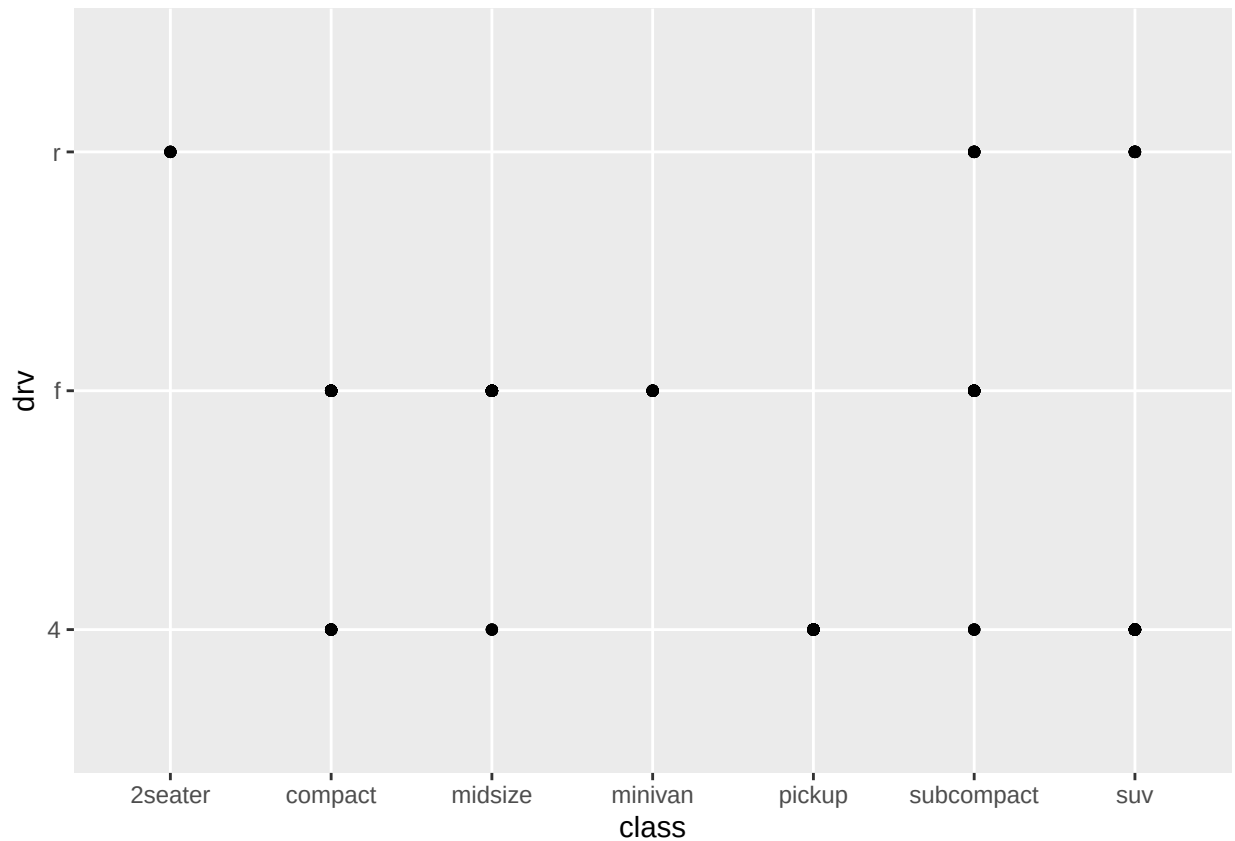
```
ggplot(data = mpg, aes(x = hwy, y = cyl)) +  
  geom_point()
```



This plot shows the highway miles per gallon (`hwy`) distributed based on the number of cylinders in the vehicle (`cyl`). We can see for example that for vehicles that have 6 cylinders, `hwy` ranges between around 14 and 29 miles per gallon.

5. What happens if you make a scatterplot of `class` vs `drv`? Why is the plot not useful?

```
ggplot(data = mpg, aes(x = class, y = drv)) +  
  geom_point()
```

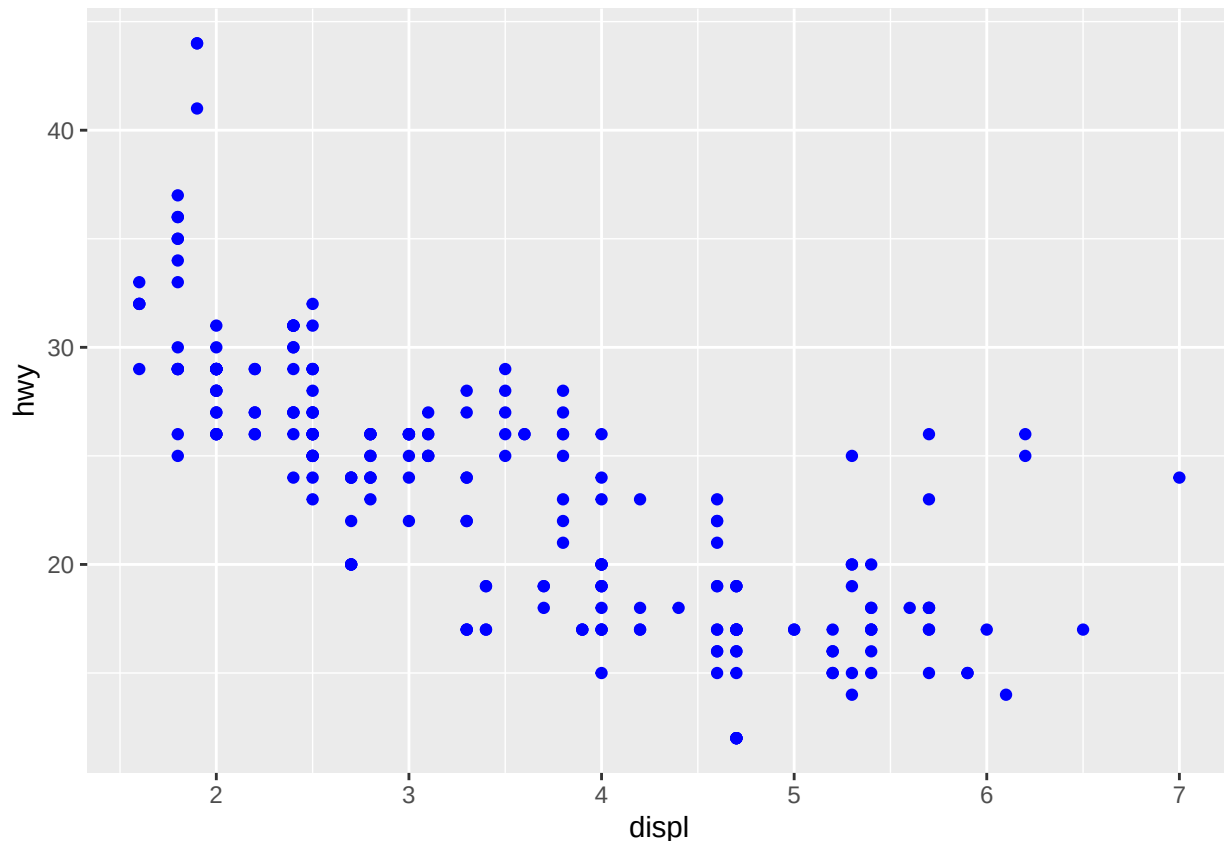


The scatterplot above does not give much useful information. This is because both class and drv are categorical variables (so both axes represent categories and not continuous variables).

3.3.1 Exercises

1. Fix the code in problem 3.3.1.1, and enter it below

```
ggplot(data = mpg) +  
  geom_point(mapping = aes(x = displ, y = hwy), color = "blue")
```



To plot in 3.3.1.1 of the book was not showing blue dots as needed since the `color="blue"` was included within the mapping argument which means it was interpreted as a mapping between a variable and a value. In other words "color" was interpreted as a categorical variable which only takes in value= "blue".

2. Which variables in `mpg` are categorical? Which variables are continuous? (Hint: type `?mpg` in the Console Panel to read the documentation for the dataset in the Help Panel). How can you see this information when you run `mpg`?

```
?mpg
mpg
```

```
## # A tibble: 234 x 11
##   manufacturer model      displ  year  cyl trans drv      cty   hwy fl      class
##   <chr>          <chr>    <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>
## 1 audi          a4         1.8  1999    4 auto~ f      18    29 p      comp~
## 2 audi          a4         1.8  1999    4 manu~ f      21    29 p      comp~
## 3 audi          a4         2    2008    4 manu~ f      20    31 p      comp~
## 4 audi          a4         2    2008    4 auto~ f      21    30 p      comp~
## 5 audi          a4         2.8  1999    6 auto~ f      16    26 p      comp~
## 6 audi          a4         2.8  1999    6 manu~ f      18    26 p      comp~
## 7 audi          a4         3.1  2008    6 auto~ f      18    27 p      comp~
## 8 audi          a4 quattro 1.8  1999    4 manu~ 4      18    26 p      comp~
## 9 audi          a4 quattro 1.8  1999    4 auto~ 4      16    25 p      comp~
## 10 audi          a4 quattro 2    2008    4 manu~ 4      20    28 p      comp~
## # i 224 more rows
```

By checking the documentation of the dataset in the Help Panel, we can conclude which variables are categorical and which are continuous by simply reading about them.

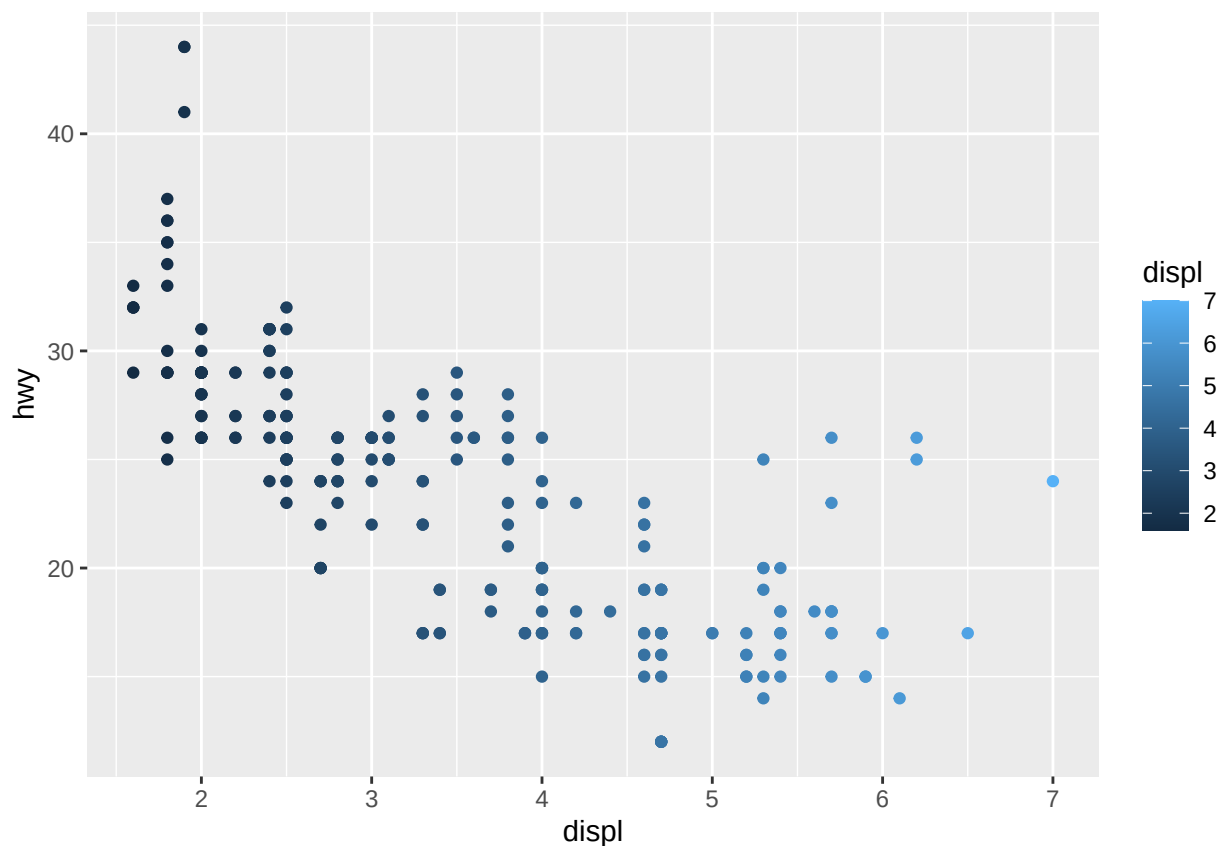
Categorical variables: manufacturer, model, trans, drv, fl, class

Continuous variables: displ, year, cyl, cty, hwy

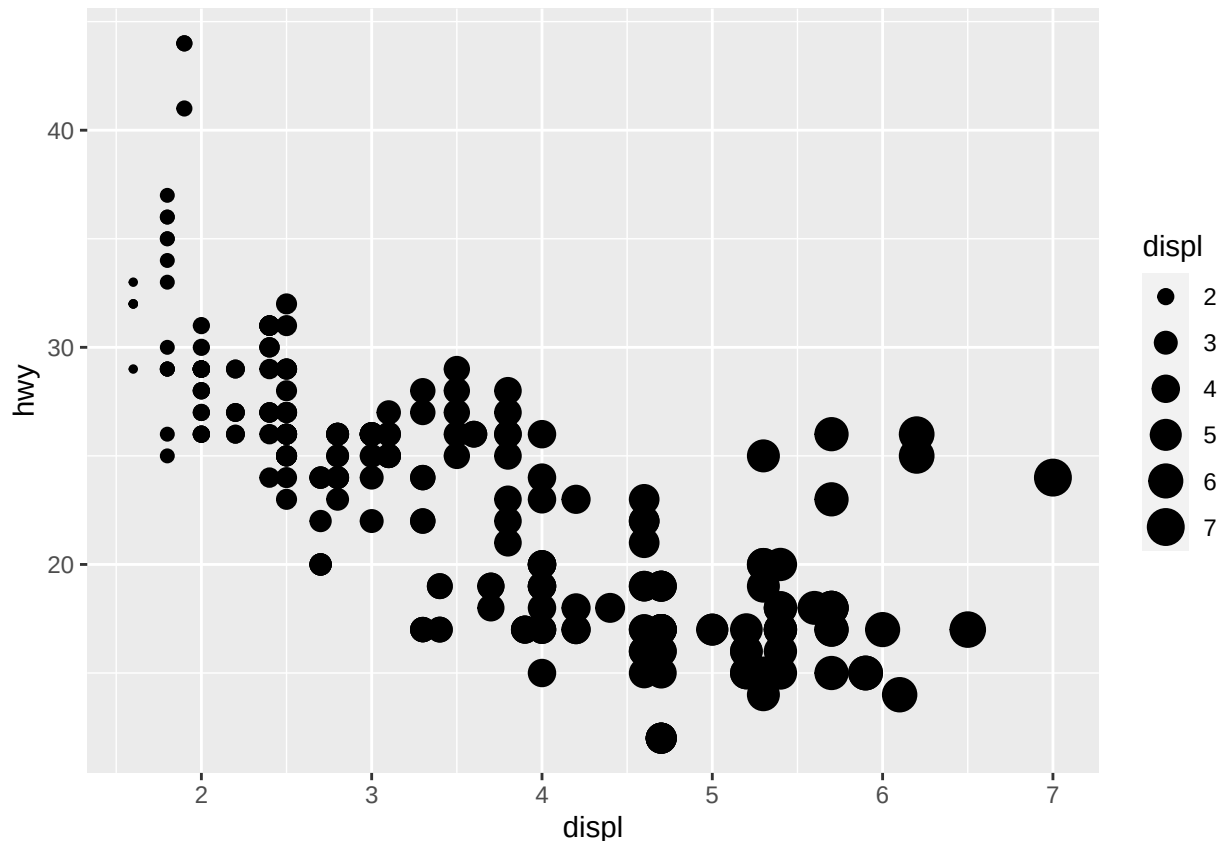
By running “mpg” we see the printed data frame, the `<...>` at the top of each column shows the type of each variable. `corresponds` to categorical variables and `and correspond` to continuous ones.

3. Using your R code for 3.3.1.1, map a continuous variable to **color**, **size**, and **shape**. How does the aesthetic **shape** behave differently for mappings to `fl` and `displ`?

```
ggplot(data = mpg) +  
  geom_point(mapping = aes(x = displ, y = hwy, color = displ))
```



```
ggplot(data = mpg) +  
  geom_point(mapping = aes(x = displ, y = hwy, size = displ))
```



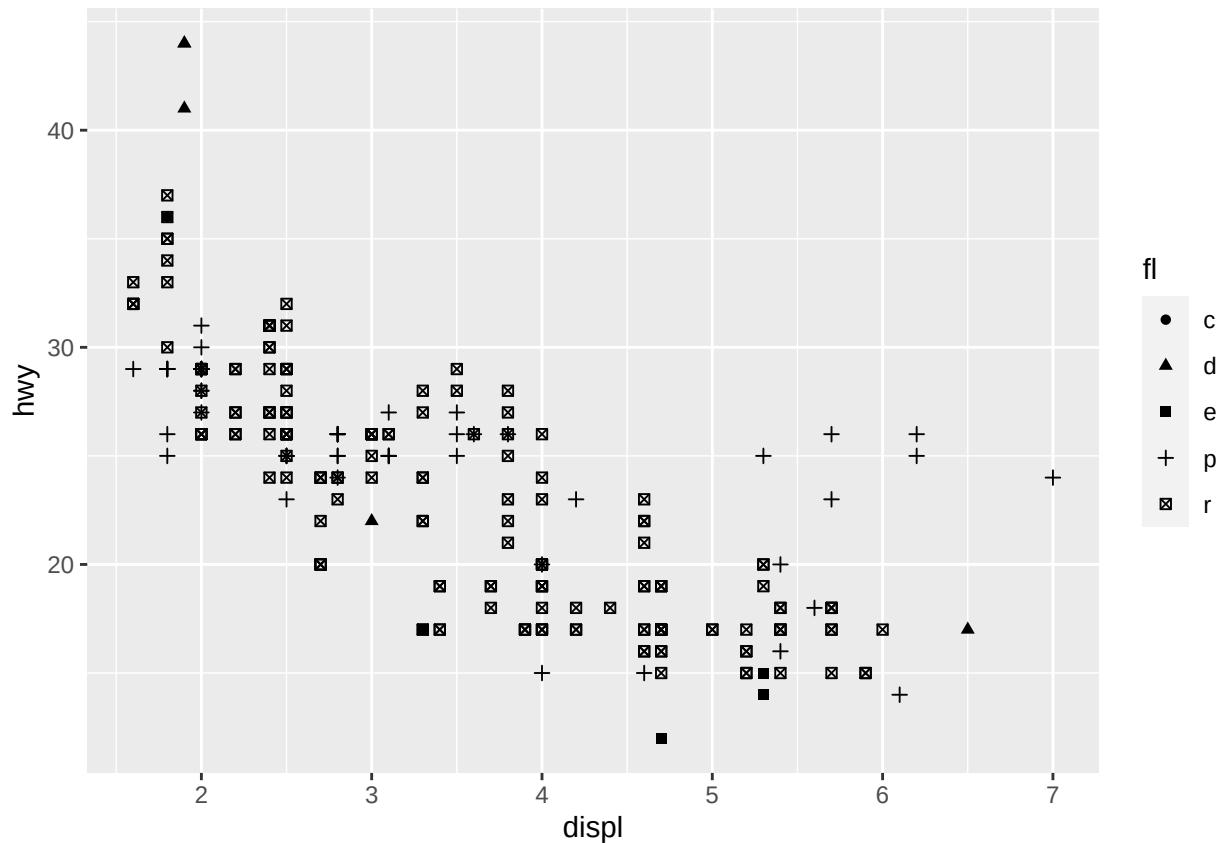
```
ggplot(data = mpg) +
  geom_point(mapping = aes(x = displ, y = hwy, shape = displ))
```

```
## Error in 'geom_point()':
## ! Problem while computing aesthetics.
## i Error occurred in the 1st layer.
## Caused by error in 'scale_f()':
## ! A continuous variable cannot be mapped to the shape aesthetic
## i choose a different aesthetic or use 'scale_shape_binned()'
```

Mapping `displ` to color or size provide us with nice plots that use color and size on points (respectively) based on `displ`. However, mapping `displ` or any continuous variable for that matter to shape gives an error since shape is usually discrete and it does not map continuous variables well (there is only a limited amount of shapes that can be given, so discrete not continuous).

Mapping `displ` to color and size produce meaningful results. Mapping `displ` to shape produces a warning. Assigning the aesthetic `shape` to `fl` produces the following.

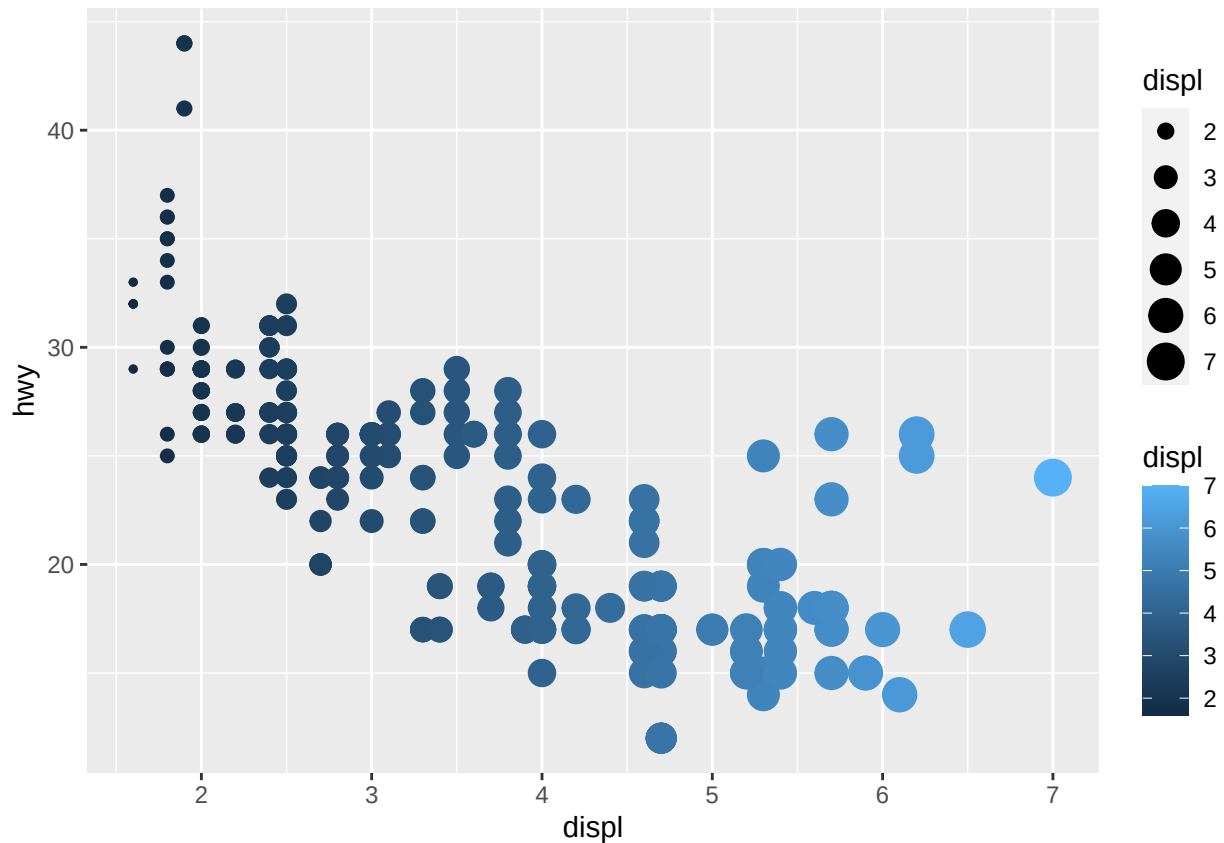
```
ggplot(data = mpg) +
  geom_point(mapping = aes(x = displ, y = hwy, shape = fl))
```



By assigning the aesthetic “shapes” to `fl` (a categorical variable), we get a nice plot which assigns different shapes to each category in `fl`. This results in different shapes for vehicles based on their fuel type.

4. What happens if you map the same variable to multiple aesthetics? Use an example from your answers to problem 3.3.1.3.

```
ggplot(data = mpg) +  
  geom_point(mapping = aes(x = displ, y = hwy, size = displ, color=displ))
```

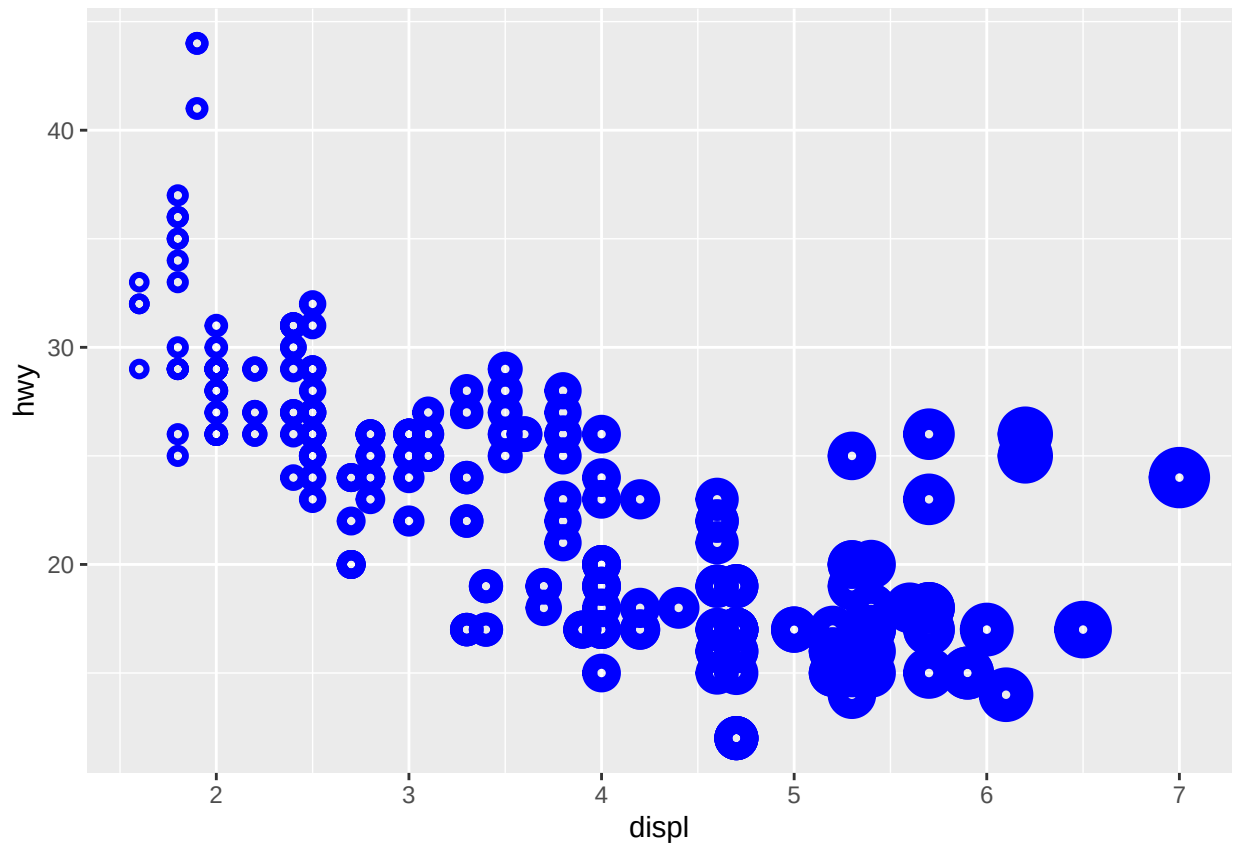



When we map the same variable to multiple aesthetics (in our case `displ` to both color and size for example), both aesthetics will scale based on the values of `displ`. This can be useful if you want to emphasize the same data in different ways (like color for grouping and size for magnitude), but in most cases it is redundant and repetitive (so not really necessarily).

5. What does the `stroke` aesthetic do? What shapes does it work with? (Hint: use `?geom_point`) Try it with `shape=21` and `stroke=displ` in your code from 3.3.1.1.

```
?geom_point
```

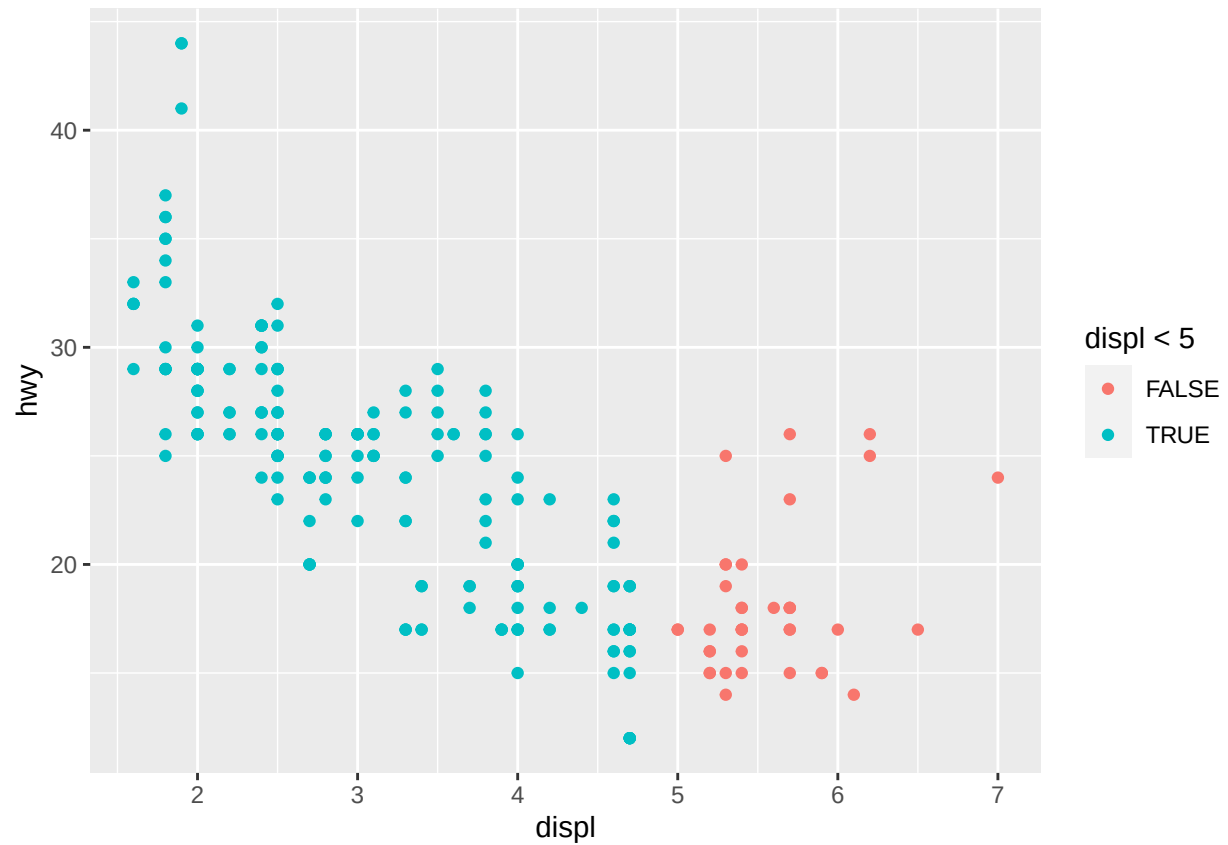
```
ggplot(data = mpg) +
  geom_point(mapping = aes(x = displ, y = hwy, stroke=displ), color = "blue", shape=21)
```



What `stroke` does is that it controls the thickness of the borders of points with certain shapes. By mapping `displ` to `stroke` we are changing the thickness of each border based on the `displ` values. In order for `stroke` to work shapes must have both an inside and an outside border (like `shape=21` or `shape=22`).

6. What happens if you map an aesthetic to something other than a variable name, like `aes(colour = displ < 5)`? Try this by modifying your code in problem 3.3.1.1.

```
ggplot(data = mpg) +  
  geom_point(mapping = aes(x = displ, y = hwy, color = displ < 5))
```



By mapping color to the logical expression `displ < 5`, it evaluates it to True or False. `ggplot2` then uses this logical condition to color code the points differently based on whether the condition is met or not.